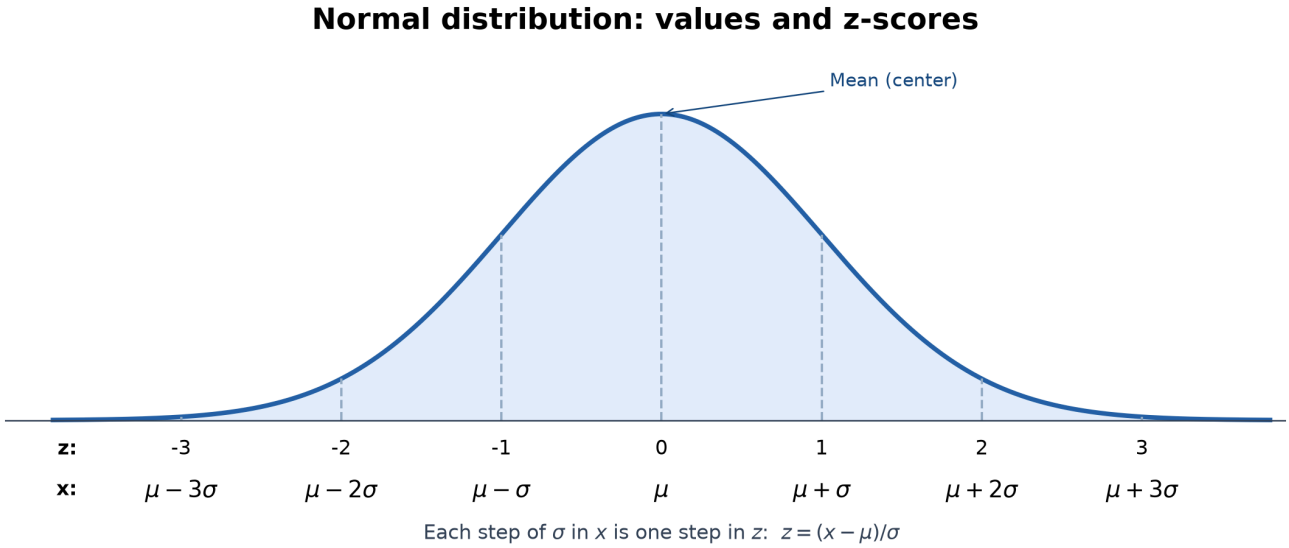
Lesson 8.1 - Standard Normal Distributions

> **Video:** Standard Normal Distribution Tables, Z Scores, Probability & Empirical Rule by The Organic Chemistry Tutor **Chapter:** 8 - Normal Probability Distributions

The normal distribution

- A **bell-shaped**, symmetric curve, written $X \sim N(\mu, \sigma)$: the first number is the **mean**, the second the **standard deviation**.
- The mean sits at the center. Each standard deviation away from it is one unit of **z-score**.



Key formulas

Purpose	Formula
z-score of a value x	$z = \dfrac{x - \mu}{\sigma}$
Value x from a z-score	$x = \mu + z\sigma$
PDF (for reference)	$f(x) = \dfrac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$

You don't need the PDF to find probabilities in an intro course. Use z-tables (or calculus, see Lesson 8.2).

- **Positive** z → the value is **above** the mean. **Negative** z → **below** the mean.

The empirical rule (68-95-99.7)

Within	Percent of the data
$\mu \pm 1\sigma$	68%
$\mu \pm 2\sigma$	95%
$\mu \pm 3\sigma$	99.7%

- $68/2 = 34\%$
- $(95 - 68)/2 = 13.5\%$
- $(99.7 - 95)/2 = 2.35\%$
- $(100 - 99.7)/2 = 0.15\%$

The total area under the curve is 1 (100%).

Example 1 - $X \sim N(50, 10)$

(a) $\mu = 50$, $\sigma = 10$.

(b) Which x has $z = 1.4$?

$$x = 50 + 1.4(10) = 64$$

(c) What is the z-score of $x = 30$?

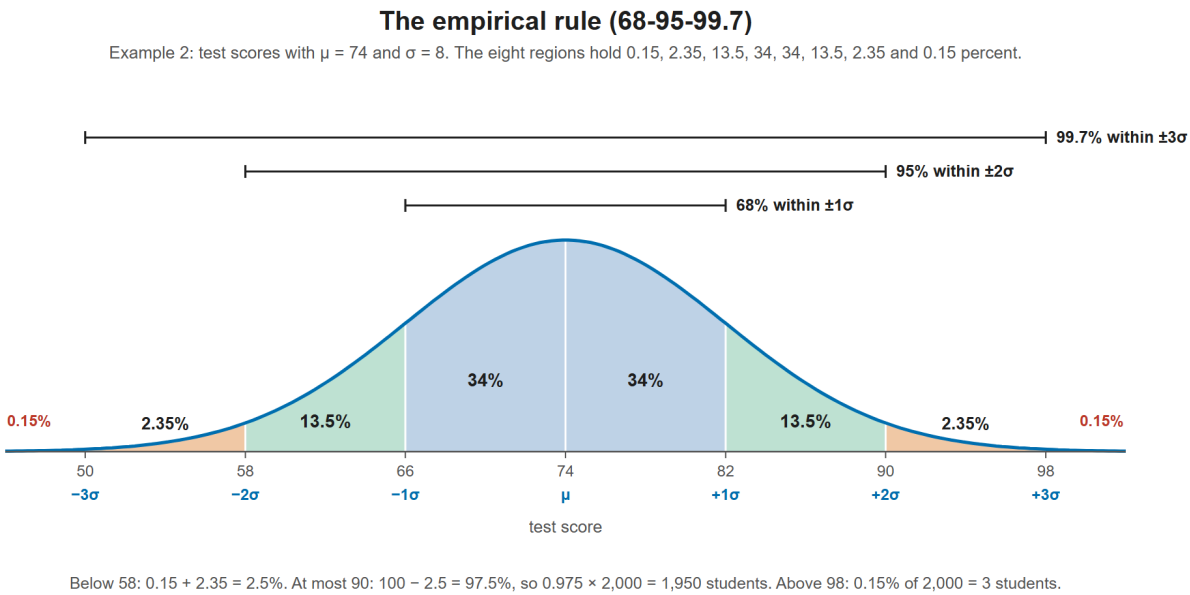
$$z = \frac{30 - 50}{10} = -2$$

(d) Negative z-scores (like −2 for 30) are left of the mean. Positive ones (like 1.4 for 64) are right of it.

Example 2 - empirical rule: test scores, $\mu = 74$, $\sigma = 8$, 2,000 students

Mark the axis: 50, 58, 66, **74**, 82, 90, 98 (from -3σ to $+3\sigma$).

Question	Regions	Answer
(a) % below 58 ($< \mu - 2\sigma$)	0.15 + 2.35	**2.5%**
(b) P(66 < X < 82) ($\mu \pm 1\sigma$)	34 + 34	**68%**
(c) **How many** scored at most 90?	everything below $+2\sigma$ = 100 − (2.35 + 0.15) = 97.5%	$0.975 \times 2000 =$ **1,950 students**
(d) % at least 66	34 + 34 + 13.5 + 2.35 + 0.15	**84%**
(e) **How many** scored more than 98?	0.15%	$0.0015 \times 2000 =$ **3 students**



Below 58: 0.15 + 2.35 = 2.5%. At most 90: 100 − 2.5 = 97.5%, so 0.975 × 2,000 = 1,950 students. Above 98: 0.15% of 2,000 = 3 students.

The empirical rule on the test-score example (μ = 74, σ = 8). Each question in Example 2 adds up the regions it covers.

Using z-tables

When z is not a whole number, such as $z = 1.56$, the empirical rule can't be used. Use a **standard normal (z) table** instead.

A z-table gives the area to the LEFT of z.

- Find the row for the first decimal (**1.5**) and the column for the second decimal (**0.06**): area = **0.94062**.
- For $z = -0.43$: row **−0.4**, column **0.03**, area = **0.33360** (33.36%).

To find	Use
$P(X < x)$	the table value directly
$P(X > x)$	$1 -$ table value
$P(a < X < b)$	table(z_b) − table(z_a)

Example 3 - IQ scores, $\mu = 100$, $\sigma = 15$

(a) P(IQ < 80)

$$z = \frac{80 - 100}{15} = -1.33 \;\Rightarrow\; \text{area} = 0.09176 \approx 9.18\%$$

(b) P(IQ > 136)

$$z = \frac{136 - 100}{15} = 2.40 \;\Rightarrow\; \text{left area} = 0.99180$$

$$P(X > 136) = 1 - 0.99180 = 0.0082 = 0.82\%$$

(The video says "eight point two" at one point; the correct value is 0.0082, which is **0.82%**.)

(c) P(95 < IQ < 110)

$$z_{110} = \frac{10}{15} \approx 0.67 \to 0.74857, \qquad z_{95} = \frac{-5}{15} \approx -0.33 \to 0.37070$$

$$P = 0.74857 - 0.37070 = 0.37787 \approx 37.8\%$$

(d) IQ at the 90th percentile (work backwards from the area)

The left area is 0.90. In the table, $z = 1.28$ gives 0.89973 and $z = 1.29$ gives 0.90147. 0.89973 is closer to 0.90, so take $z = 1.28$.

$$x = 100 + 1.28(15) = 119.2$$

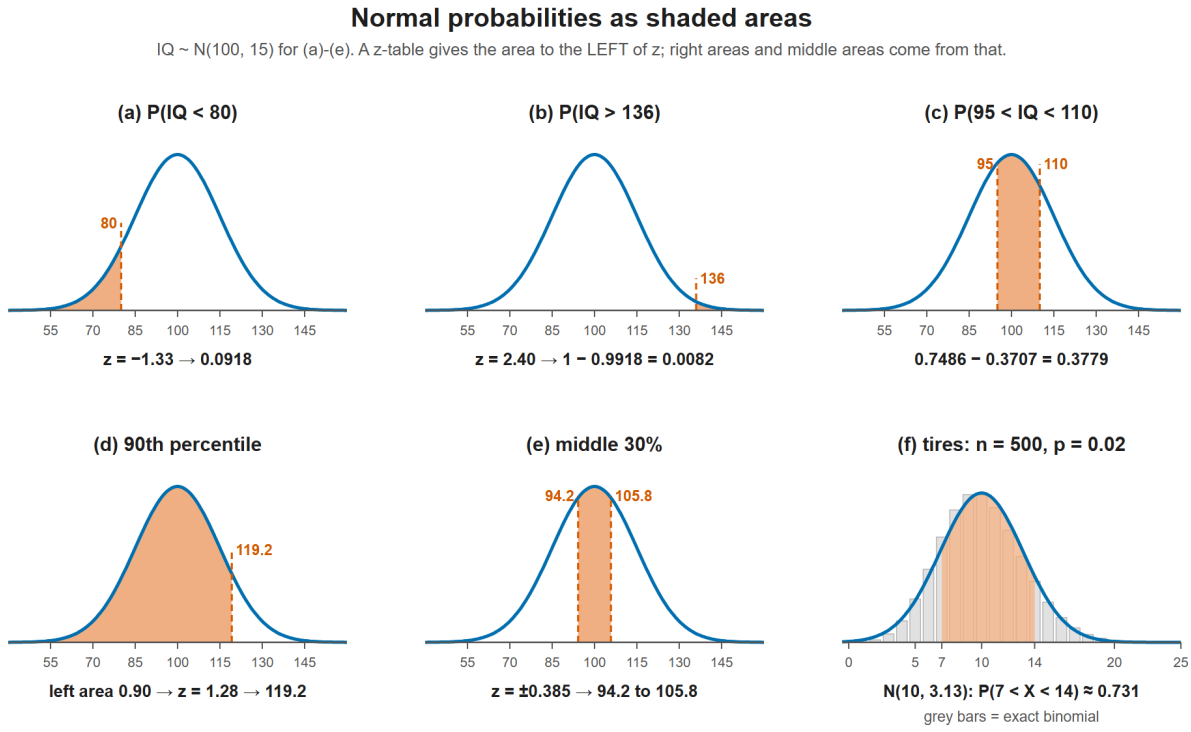
(e) Between which two IQs are the middle 30%?

The middle 30% leaves 15% on each side of the mean, so it runs from the **35th** to the **65th** percentile.

- For an area of 0.65: $z = 0.38$ gives 0.64803 and $z = 0.39$ gives 0.65173. 0.65 is about halfway, so $z \approx 0.385$.
- By symmetry, the 35th percentile is $z \approx -0.385$.

$$x_2 = 100 + 0.385(15) \approx 105.8, \qquad x_1 = 100 - 0.385(15) \approx 94.2$$

The middle 30% of IQs lie between about **94.2 and 105.8**.



Each IQ question as a shaded area. A left area comes straight from the z-table, a right area is 1 minus it, and a middle area is the difference of two left areas. For percentiles, start from the area and work back to x. In (f) the normal curve closely follows the exact binomial bars.

Example 4 - normal approximation to a binomial (defective tires)

2% of tires are defective, and a sample of $n = 500$ is checked. So $p = 0.02$, $q = 0.98$.

(a) Mean and standard deviation

$$\mu = np = 500(0.02) = 10, \qquad \sigma = \sqrt{npq} = \sqrt{500(0.02)(0.98)} = \sqrt{9.8} \approx 3.13$$

(b) P(fewer than 8 defective)

$$z = \frac{8 - 10}{3.13} \approx -0.64 \;\Rightarrow\; P \approx 0.2611 = 26.1\%$$

(c) P(more than 15 defective)

$$z = \frac{15 - 10}{3.13} \approx 1.60 \;\Rightarrow\; \text{left} = 0.9452,\; P = 1 - 0.9452 = 0.0548 = 5.48\%$$

(d) P(between 7 and 14 defective)

$$z_{14} = \frac{4}{3.13} \approx 1.28 \to 0.89973, \qquad z_{7} = \frac{-3}{3.13} \approx -0.96 \to 0.16853$$

$$P = 0.89973 - 0.16853 = 0.7312 = 73.1\%$$

> **Note:** the video uses the normal curve directly on the counts. Many textbooks add a **continuity correction** of ±0.5 (e.g. "fewer than 8" → $x < 7.5$) when approximating a discrete binomial with the continuous normal, which gives slightly different, more accurate answers.

Key takeaways

- $z = (x - \mu)/\sigma$ and $x = \mu + z\sigma$.
- Empirical rule: 68 / 95 / 99.7, which splits into 34 / 13.5 / 2.35 / 0.15 per region.
- A z-table gives the **left** area. Right area = 1 − left. Between = difference of two left areas.
- For percentiles, look up the **area** in the table to find z, then convert to x.

Lesson 8.2 - Standard Normal Distribution with Calculus

> **Video:** Normal Distribution & Probability Problems by The Organic Chemistry Tutor **Chapter:** 8 - Normal Probability Distributions

More precise empirical-rule percentages

The bell curve runs from $-\infty$ to $+\infty$ with total area 1 (100%). Each half is 50%.

Within	Percent
$\mu \pm 1\sigma$	68.27%
$\mu \pm 2\sigma$	95.45%
$\mu \pm 3\sigma$	99.73%

Per region, using symmetry:

```
0.135% | 2.14% | 13.59% | 34.13% | 34.13% | 13.59% | 2.14% | 0.135%
    μ-3σ     μ-2σ      μ-σ         μ        μ+σ      μ+2σ     μ+3σ
```

- $68.27/2 \approx 34.13$
- $(95.45 - 68.27)/2 \approx 13.59$
- $(99.73 - 95.45)/2 = 2.14$
- $(100 - 99.73)/2 = 0.135$

The exact method: integrate the PDF

$$P(a < X < b) = \int_a^b \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}\, dx$$

There is no simple antiderivative, so evaluate it with a **graphing calculator** or an online tool such as a definite-integral calculator (e.g. Wolfram|Alpha).

Typing tips: put the whole exponent in parentheses and include the negative sign. For $\mu = 75$, $\sigma = 7$:

```
e^(-((x-75)^2)/(2*7^2)) / (7*sqrt(2*pi))
```

For "greater than" questions, the upper limit is $+\infty$, not the maximum score. Integrating up to 100 underestimates the answer. Use a very large number (1,000; 100,000), or ∞ if the tool accepts it, and check that the result has stopped changing.

Example - physics test scores

800 students; scores are normal with $\mu = 75$, $\sigma = 7$.

Axis marks: 54, 61, 68, **75**, 82, 89, 96 (from -3σ to $+3\sigma$).

(a) What percentage scored between 68 and 82? ($\pm 1\sigma$)

- Empirical rule: $34.134 + 34.134 = 68.27\%$
- Integral from 68 to 82: $0.682689 \rightarrow$ **68.27%** ✓

(b) About how many students scored between 61 and 89? ($\pm 2\sigma$)

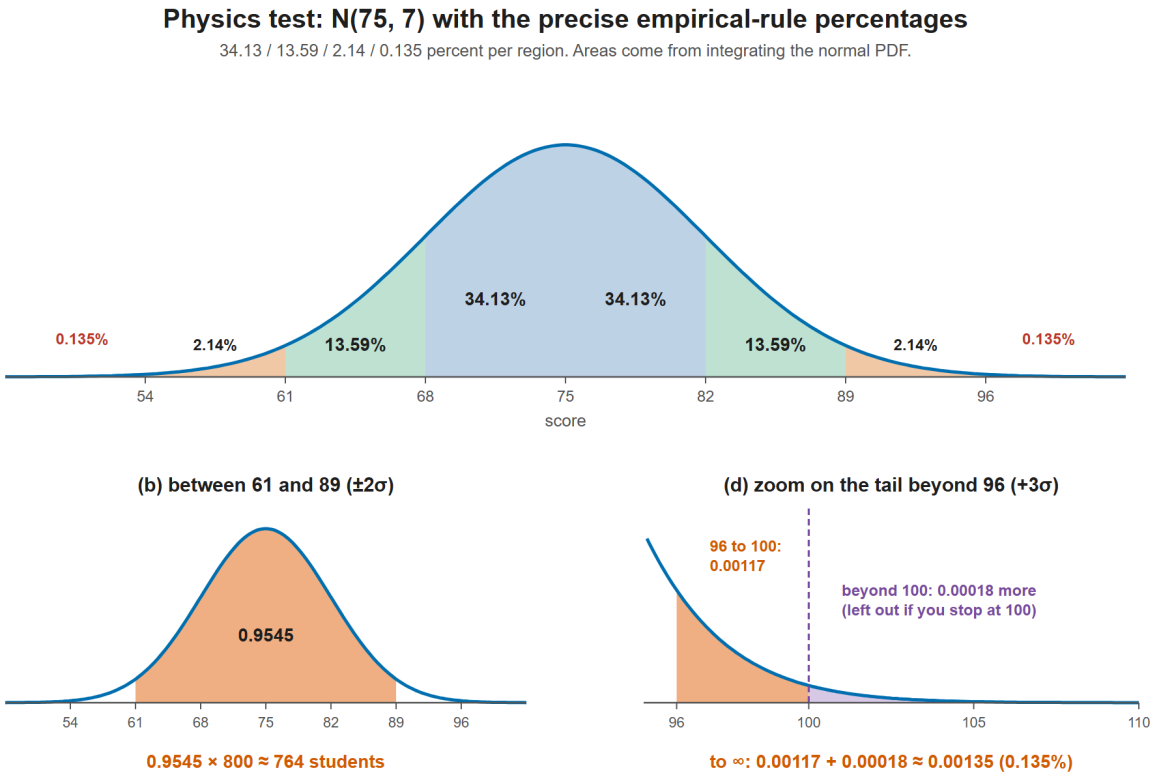
- Empirical rule: $13.59 + 34.13 + 34.13 + 13.59 \approx 95.45\%$
- Integral from 61 to 89: 0.954500
- Number of students: $0.9545 \times 800 \approx 763.6 \rightarrow$ **about 764 students**

(c) P(score between 54 and 75) (-3σ to the mean)

- Empirical rule: $2.14 + 13.59 + 34.13 \approx 49.87\%$
- Integral from 54 to 75: $0.498650 \rightarrow$ **49.87%**

(d) About how many students scored 96 or more? (beyond $+3\sigma$)

- Empirical rule: **0.135%**
- Integral from 96 to 100: 0.00117. This is **too small**, because the curve continues past 100 (think of extra credit).
- Integral from 96 to 1,000 (effectively ∞): $0.0013499 \approx 0.135\%$ ✓
- Number of students: $0.0013499 \times 800 \approx 1.08 \rightarrow$ **about 1 student**



Top: the more precise region percentages. Bottom right: integrating only up to 100 captures the orange part (0.00117) but misses the purple part beyond 100. Integrating to infinity gives the full 0.00135.

Key takeaways

- More precise empirical-rule values: **68.27 / 95.45 / 99.73**, which split into 34.13 / 13.59 / 2.14 / 0.135 per region.
- The exact probability is the integral of the normal PDF between the limits. Use technology.
- For tails, integrate out to **infinity**, using a very large number if needed.
- To get a **count**, multiply the probability by the group size and round to a whole number.

Lesson 8.3 - Law of Large Numbers

> **Video:** Law of Large Numbers by The Organic Chemistry Tutor **Chapter:** 8 - Normal Probability Distributions

Statement

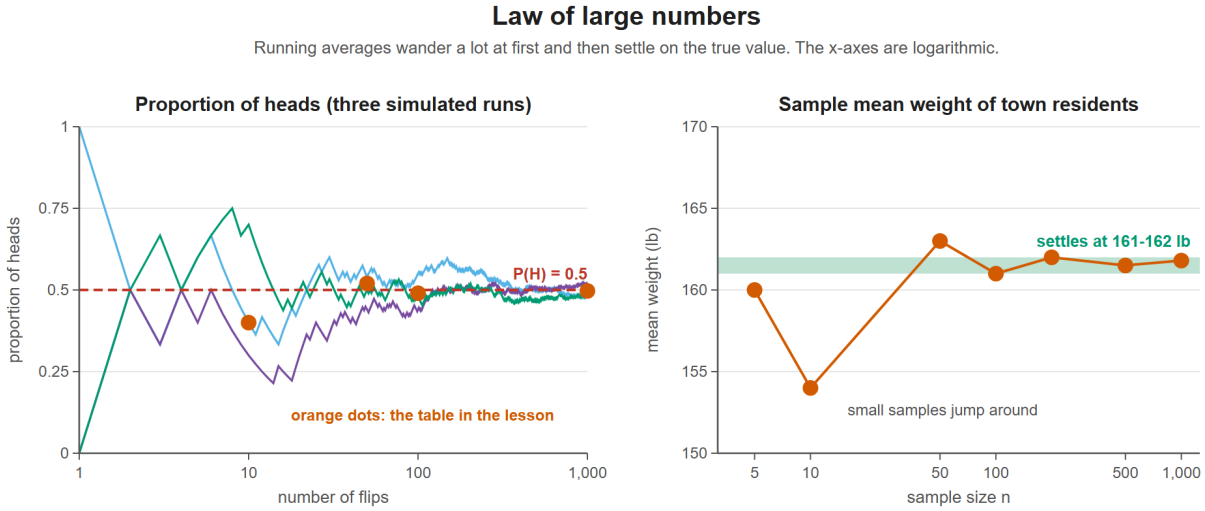
> If an experiment is repeated many times, the **observed average** (or observed proportion) gets **closer and closer to the true, theoretical value** as the number of trials increases.

Example 1 - flipping coins

The theoretical probability of heads is $P(H) = \frac{1}{2} = 0.5$. A few trials can stray far from 0.5, but many trials land close to it:

Number of flips	Heads observed (example)	Observed proportion
10	4	0.40
50	26	0.52
100	49	0.49
1,000	497	0.497

As the number of flips grows, the observed proportion **approaches 0.5**.



Left: three simulated runs of 1,000 coin flips. Each wanders early on, then all three close in on 0.5. Right: the sample mean weights from the lesson, which jump around for small samples and settle near 161-162 lb.

Example 2 - estimating an average weight

A town has 20,000 residents. Instead of weighing everyone, take samples of increasing size:

Sample size n	Sample mean weight (example)
5	160 lb
10	154 lb
50	163 lb
100	161 lb
200	162 lb
500	161.5 lb
1,000	161.8 lb

- **Small samples** jump around, because one unusually heavy (230 lb) or light (110 lb) person has a big effect.
- **Large samples** settle down. Here they converge to about **161-162 lb**, which estimates the population mean.

Key takeaways

- More trials or a bigger sample → the observed average gets closer to the true average.
- This is why large samples give reliable estimates of population **parameters**.
- The law says nothing about any single trial. Even after many heads in a row, the next flip is still 50/50.

Lesson 8.4 - The Central Limit Theorem

> **Video:** Central Limit Theorem - Sampling Distribution of Sample Means by The Organic Chemistry Tutor
> **Chapter:** 8 - Normal Probability Distributions

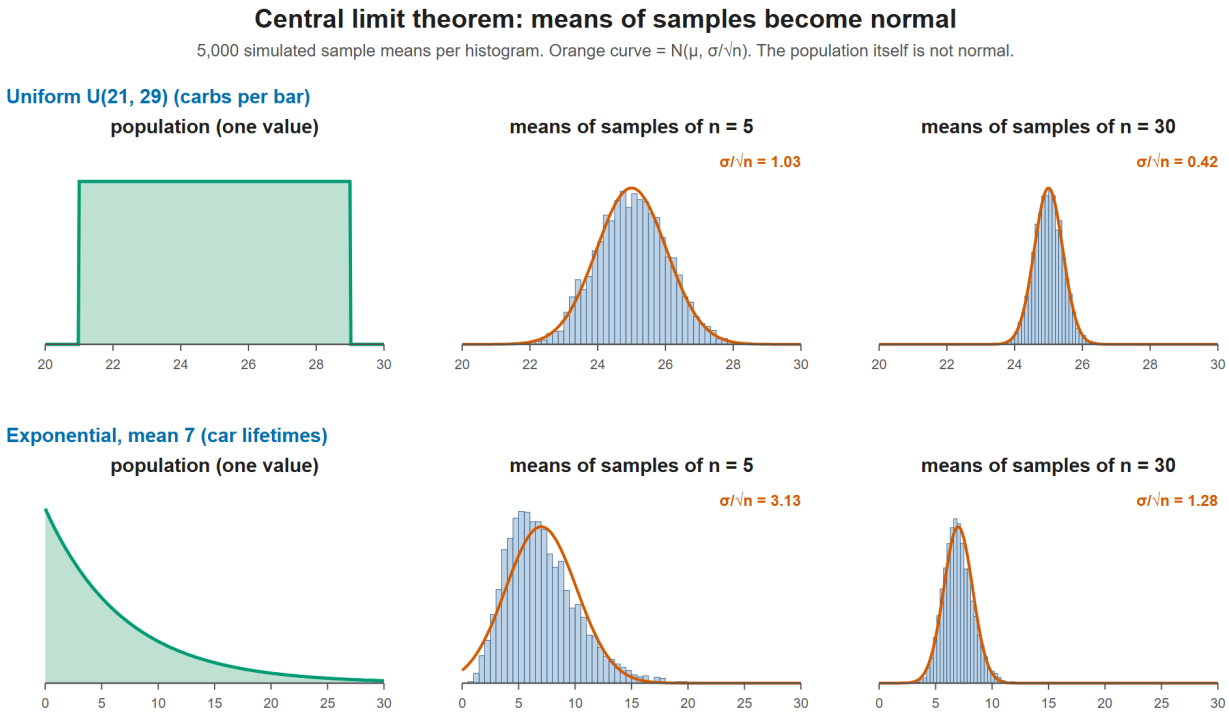
The idea

1. Take many samples of size n from a population.
2. Compute the mean $\bar{x}$ of each sample.
3. Plot all those sample means in a histogram.

That histogram is the **sampling distribution of the sample means**, and it is **approximately normal**:

Population shape	Sampling distribution of $\bar{x}$
Normal	Normal for **any** n
Anything else (uniform, exponential, skewed, irregular, ...)	Approximately normal when n is large enough. $n \geq 30$ is the usual rule of thumb

Example: in a town of 100,000, take repeated samples of 30 people and record each sample's mean age (41.8, 39.6, 40.5, ...). Those means form a bell curve even if individual ages do not.



Larger n: the histogram becomes more bell-shaped AND narrower, always centered on the population mean μ.

Neither population is normal, yet the sample means pile up into a bell shape. With n = 30 they match the normal curve N(μ, σ/√n) closely, even for the skewed exponential population, where n = 5 is still visibly lopsided.

Symbols

Symbol	Meaning
μ	Population mean
σ	Population standard deviation
$\bar{x}$	Mean of **one** sample
s	Standard deviation of **one** sample
$\mu_{\bar{x}}$	Mean of the **sampling distribution**: $\mu_{\bar{x}} = \mu$
$\sigma_{\bar{x}}$	SD of the sampling distribution, the **standard error**: $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$
n	Sample size

Law of large numbers link: as n grows, $\bar{x}$ gets closer to μ. For example, mean ages from samples of 10, 50, 100 and 1,000 people might be 32, 38.5, 35.3 and 36.1, approaching the true 36.

Larger n → narrower curve

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

n is in the denominator, so **a larger sample gives a smaller standard error**. The sampling distribution becomes **taller and narrower**: $n = 50$ is narrower than $n = 30$, which is narrower than $n = 10$.

---

z-scores: individuals vs means vs sums

Question about...	Distribution	z-score
One value x	$X \sim N(\mu, \sigma)$	$z = \frac{x - \mu}{\sigma}$
The **mean** of n values	$\bar{X} \sim N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$	$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$
The **sum** of n values	$\sum X \sim N(n\mu, \sqrt{n}\,\sigma)$	$z = \frac{\sum x - n\mu}{\sqrt{n}\,\sigma}$

Then use the z-table (left area); right area = 1 − left area. To go from a percentile back to a value: $\bar{x} = \mu + z\,\frac{\sigma}{\sqrt{n}}$.

Reminders from earlier lessons

- **Uniform** $U(a, b)$: $f(x) = \frac{1}{b-a}$, $\mu = \frac{a+b}{2}$, $\sigma = \frac{b-a}{\sqrt{12}}$, probability = base × height.
- **Exponential**: $f(x) = \lambda e^{-\lambda x}$, $\lambda = 1/\mu$, $\sigma = \mu$, $P(X < x) = 1 - e^{-\lambda x}$.

Example 1 - entrance exam scores: normal, $\mu = 74$, $\sigma = 6.8$

(a) One student scores below 65 ($n = 1$, so use the population distribution)

$$z = \frac{65 - 74}{6.8} \approx -1.32 \;\Rightarrow\; P = 0.09342 \approx 9.34\%$$

(b) The mean of 50 students is above 75

$$\sigma_{\bar{x}} = \frac{6.8}{\sqrt{50}} \approx 0.9617, \qquad z = \frac{75 - 74}{0.9617} \approx 1.04$$

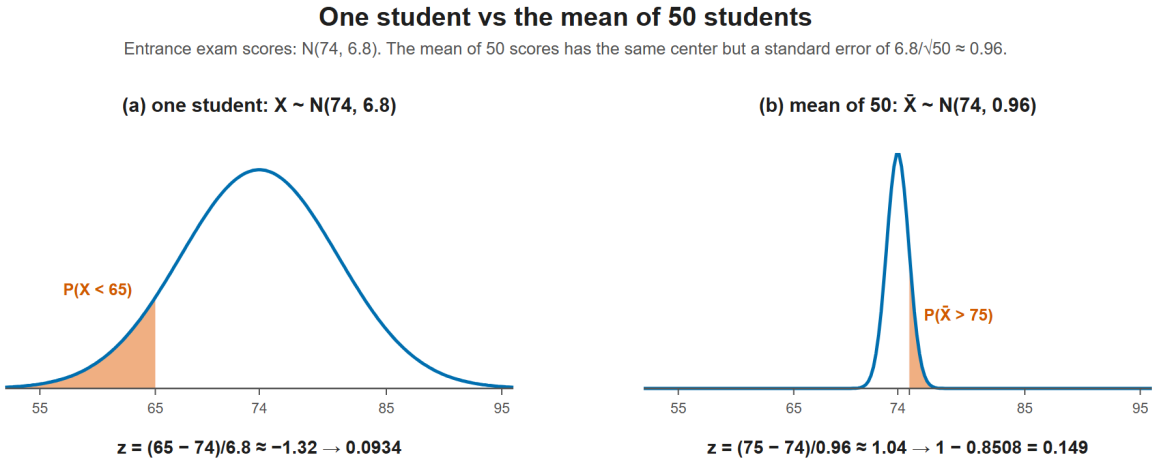
$$P(\bar{X} > 75) = 1 - 0.85083 = 0.14917 \approx 14.9\%$$

(c) Distribution of the mean of 50 scores

$$\bar{X} \sim N(74,\ 0.9617)$$

(d) 80th percentile of the mean score A left area of 0.80 gives $z \approx 0.84$ (table value 0.79955):

$$\bar{x} = 74 + 0.84(0.9617) \approx 74.81$$



Example 1 on a shared axis. A question about one student uses σ = 6.8. A question about the mean of 50 students uses the much smaller standard error 6.8/√50 ≈ 0.96.

Example 2 - carbs in a snack bar: uniform from 21 g to 29 g, $n = 100$

(a) One bar

$$X \sim U(21, 29), \qquad \mu = \frac{21 + 29}{2} = 25, \qquad \sigma = \frac{29 - 21}{\sqrt{12}} \approx 2.3094$$

(b) Mean of 100 bars. By the CLT ($n \geq 30$) it is **normal**, even though one bar is uniform:

$$\bar{X} \sim N\left(25,\ \frac{2.3094}{\sqrt{100}}\right) = N(25,\ 0.23094)$$

(c) One bar has 24-26 g (uniform, so base × height)

$$P(24 < X < 26) = (26 - 24) \times \frac{1}{8} = 0.25$$

(d) The mean of 100 bars is between 24.9 g and 25.1 g (normal)

$$z = \frac{\pm 0.1}{0.23094} \approx \pm 0.43 \;\Rightarrow\; P = 0.66640 - 0.33360 = 0.3328$$

(e) Distribution of the total carbs in 100 bars

$$\sum X \sim N\big(n\mu,\ \sqrt{n}\,\sigma\big) = N\big(100 \cdot 25,\ 10 \cdot 2.3094\big) = N(2500,\ 23.094)$$

(f) The total is more than 2,540 g

$$z = \frac{2540 - 2500}{23.094} \approx 1.73 \ \Rightarrow\ P = 1 - 0.95818 = 0.04182 \approx 4.2\%$$

Example 3 - car lifetimes: exponential with mean 7 years, $n = 40$ cars

(a) Rate and SD (one car)

$$\lambda = \frac{1}{7} \text{ per year}, \qquad \sigma = \mu = 7 \text{ years}$$

(b) Distribution of the mean lifetime of 40 cars (CLT: normal)

$$\sigma_{\bar{x}} = \frac{7}{\sqrt{40}} \approx 1.1068, \qquad \bar{X} \sim N(7,\ 1.1068)$$

(c) P(sample mean < 6.5 years)

$$z = \frac{6.5 - 7}{1.1068} \approx -0.45 \ \Rightarrow\ P = 0.32636 \approx 32.6\%$$

(d) IQR of the sample mean

- Q_1 (left area 0.25): the table gives $z = -0.67 \to 0.2514$ and $z = -0.68 \to 0.2483$, so use $z \approx -0.675$.

$$Q_1 = 7 - 0.675(1.1068) \approx 6.253$$

- Q_3 (left area 0.75): $z \approx +0.675$

$$Q_3 = 7 + 0.675(1.1068) \approx 7.747$$

- $$\text{IQR} = 7.747 - 6.253 \approx 1.494 \text{ years}$$

Key takeaways

- Sample means are approximately **normal** for $n \geq 30$, whatever the population's shape (exactly normal if the population is normal).
- $\mu_{\bar{x}} = \mu$ and $\sigma_{\bar{x}} = \sigma/\sqrt{n}$, so a bigger n gives a smaller spread.
- Decide whether the question is about **one** value (use σ and the population's own shape), a **mean** (use $\sigma/\sqrt{n}$, normal), or a **sum** (use $n\mu$ and $\sqrt{n}\,\sigma$, normal).

Lesson 8.5 - The Standard Error of the Mean

> **Video:** Standard Error of the Mean by The Organic Chemistry Tutor **Chapter:** 8 - Normal Probability Distributions

Formula

The **standard error of the mean (SE)** is the standard deviation of the sampling distribution of $\bar{x}$. It measures how much sample means vary from sample to sample.

You know	Standard error
The **sample** SD s	$\text{SE} = \dfrac{s}{\sqrt{n}}$
The **population** SD σ	$\sigma_{\bar{x}} = \dfrac{\sigma}{\sqrt{n}}$

Example 1 - from sample data

The ages of 5 randomly selected college students are 18, 21, 19, 20, 26.

(a) Sample standard deviation

Mean: $\bar{x} = \dfrac{104}{5} = 20.8$

x	$x - 20.8$	squared
18	−2.8	7.84
21	0.2	0.04
19	−1.8	3.24
20	−0.8	0.64
26	5.2	27.04
Sum		**38.80**

$$s = \sqrt{\frac{38.8}{5-1}} = \sqrt{9.7} \approx 3.114$$

(b) Standard error

$$\text{SE} = \frac{3.114}{\sqrt{5}} \approx 1.393$$

Example 2 - from the population SD

At a university, student ages have mean 20.5 and **population** SD $\sigma = 0.8$.

(a) Samples of $n = 25$

$$\sigma_{\bar{x}} = \frac{0.8}{\sqrt{25}} = \frac{0.8}{5} = 0.16$$

(b) Samples of $n = 100$

$$\sigma_{\bar{x}} = \frac{0.8}{\sqrt{100}} = \frac{0.8}{10} = 0.08$$

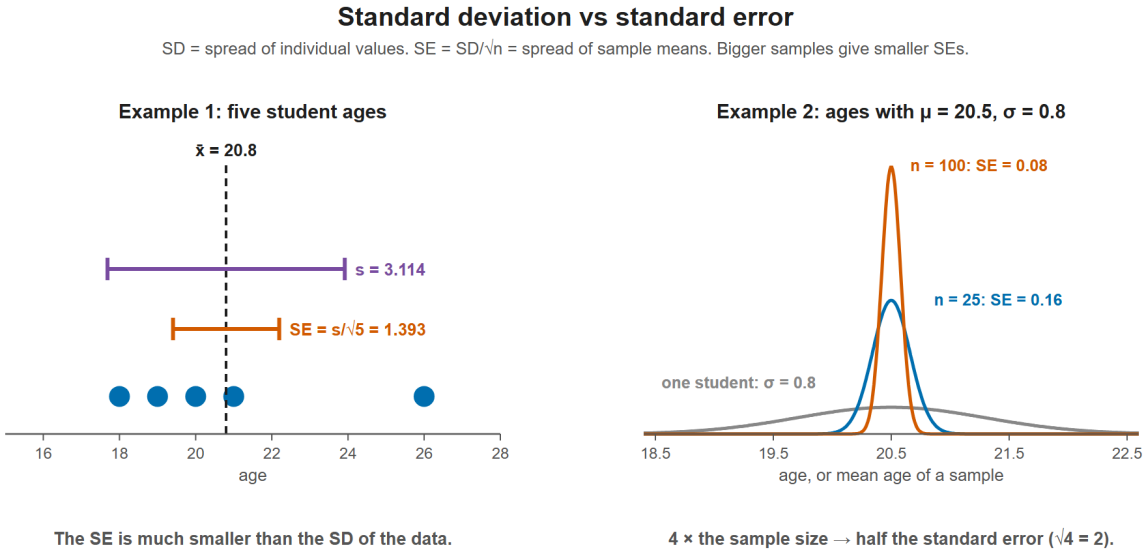
Quadrupling n halves the standard error, because $\sqrt{4} = 2$.

What it looks like

Plot the means of many samples to get the **sampling distribution of the mean**, a bell curve centered at μ.

- With $n = 25$ it is wider and shorter.
- With $n = 50$ (or 100) it is **narrower and taller**.

A smaller SE means the sample means cluster more tightly around μ, so any one sample mean is a **more accurate** estimate.



Left: the SD describes how spread out the five ages are, while the SE describes how much the mean of five ages would vary. Right: going from n = 25 to n = 100 halves the SE, so the curve of sample means gets twice as narrow.

Key takeaways

- SE = SD ÷ $\sqrt{n}$. Use s from a sample or σ from the population.
- The SD describes spread among **individual values**. The SE describes spread among **sample means**.
- SE ∝ $1/\sqrt{n}$: a larger sample gives a smaller SE and a more precise estimate.